

Dilara Caglar

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EDUCATION

Carnegie Mellon University

Anticipated Graduation - **May 2028**

B.S. Electrical and Computer Engineering, Additional Major in Engineering Design Innovation & Entrepreneurship

Relevant Coursework: *Computer Systems, Digital Systems Design, Analog Circuits, Signals & Systems, Data Structures, Linear Algebra, Differential Equations, Robotics, Computer Architecture, PCB Design*

RELEVANT EXPERIENCE

Undergraduate Researcher | Nexus Research Group | Carnegie Mellon University

January 2026 - Present

Advisor: Tathagata Srimani

- Implemented monolithic 3D integration using CNFET logic and RRAM memory, improving logic density $2.5\times$ in an AI accelerator SoC architecture
- Achieved 2.0 GHz timing closure under strict PPA constraints using congestion-aware placement, buffering, and static timing analysis (STA)
- Completed ASIC RTL-to-GDSII physical design flow including synthesis, floorplanning, place-and-route, clock tree synthesis, and physical verification
- Built Python-based automation pipelines for RTL verification and validation, reducing verification turnaround time 40% through automated test generation and log analysis
- Developed internal developer tooling for ASIC design workflows, including Python scripts for simulation analysis, verification data processing, and debugging
- Designed and validated high-speed interfaces (PCIe Gen4, DDR4, 10+ Gbps SerDes) through system-level hardware testing and instrumentation
- Developed a multi-FPGA prototyping platform (4 FPGAs) for scalable system verification and hardware bring-up of accelerator subsystems

Hardware Architecture Research Intern | SAFARI Research Group | ETH Zürich

May 2025 - December 2025

Advisor: Prof. Onur Mutlu

(Onsite Summer 2025, Remote thereafter)

- Reduced memory access overhead by 40% in LPDDR5 systems by developing hardware/software co-design techniques for homomorphic encryption workloads, improving system architecture performance
- Increased memory bandwidth utilization and data movement efficiency by extending CIPHERMATCH framework with Ramulator 2.0 and implementing RowClone-based mechanisms, enabling faster architecture-level validation and analysis

Teaching Assistant | Electrical & Computer Engineering | Carnegie Mellon University

August 2025 - December 2025

- Mentored 200+ students for circuit design, testing, and debugging in analog circuit labs and hardware validation techniques

PROJECT EXPERIENCE

Computer Systems

Fall 2025

- Improved system performance $3\times$ using cache-aware data structures, SIMD vectorization, and multithreaded optimization
- Developed concurrent systems in C/C++, including custom memory allocators, synchronization primitives, and thread-safe data structures

Embedded Systems: GPU-Accelerated Simulation

Spring 2026

- Developed a GPU-accelerated simulation framework using NVIDIA Isaac Sim for automated evaluation of compute-efficient sensor configurations
- Implemented parallel ray-casting kernels in C++/CUDA for LiDAR simulation, enabling evaluation of multi-modal pipelines
- Designed software experimentation pipelines for data-driven optimization of sensing architectures

Electronic Devices and Analog Circuits

Fall 2025

- Designed and validated CMOS and mixed-signal circuits (ring oscillators, comparators, charge pumps, SMPS) using lab instrumentation and embedded control systems

Build18 Hackathon

January 2025

- Built a real-time embedded vision system using Arduino and PyTorch with closed-loop PID control, with <100 ms latency

LEADERSHIP, ACTIVITIES & HONORS

CMU College of Engineering Student Council | *Secretary*

August 2025 - Present

CMU College of Engineering: First Year Advisory Board | *Head of Marketing*

August 2024 - May 2025

SKILLS

Hardware: ASIC Design, SoC Design, Hardware Architecture, RTL-to-GDSII, Physical Design, Synthesis, Place and Route, Clock Synthesis, Static Timing Analysis, SI/PI Analysis, Power Analysis, PDN Design, PCB Design, DRAM, PIM, Mixed-Signal Circuits

Software: C/C++, Python, Verilog/SystemVerilog, VHDL, Assembly, Java

Tool & Methods: Altium, LTspice, VSCode, GDB, Valgrind, Ramulator 2.0, Autodesk Fusion